

Luca Choteborsky

Production Engineer at Meta — M.Eng Computer Science, University of Cambridge
Github LinkedIn

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PROFILE

Production Engineer at Meta with a background in large-scale datacenter networks, AI training infrastructure, and systems software. M.Eng Computer Science from the University of Cambridge. Experienced in monitoring and maintaining fleets of 100,000+ network switches, large-scale data processing, and developer tooling. Interested in specialising in compiler/systems, AI infrastructure, and ML platforms.

ACADEMICS

University of Cambridge

October 2024 – June 2025

M.Eng Computer Science

- Principles of Machine Learning Systems: Studied and implemented algorithms and system techniques in Pytorch that are used to support the training and inference of machine learning models under a spectrum of computing systems.
- Large-scale data processing and optimisation: Provided hands-on experience with data flow programming using Google TensorFlow and cluster computing, as well as a academic knowledge on the next generation of distributed systems.
- Dissertation: built and evaluated a federated Kubernetes scheduler (Go) using online FPCA-based load prediction and distributed gRPC aggregation across cluster nodes, outperforming the default kube-scheduler on utilisation balance and job completion time across CPU-intensive and ML inference workloads.

University of Cambridge

October 2021 – June 2024

BA Computer Science

1st Class

- Computer Networking and Principles of Communication: In-depth understanding of routing, switching and network protocols including DNS, DHCP, TCP/IP.
- Advanced Operating Systems: Learned systems-analysis methodology and perform tracing and performance profiling experiments using DTrace on Linux I/O, IPC, and network-stack implementations.

A-Levels

August 2021

Dame Alice Owens School

- Achieved 4× A*s in Mathematics, Further Mathematics, Computer Science and Physics
- Awarded the **Computer Science School Prize** for academic excellence

TECHNICAL SKILLS

Programming Languages	Rust, C++, C, C#, Python, Go, Java, SQL, OCaml, Bash, Perl, L ^A T _E X
Framework & Tools	Kubernetes, Docker, React, JavaScript, TypeScript, Google Cloud VMs, AWS
Networking & Systems	BGP, DHCP, ZTP, Thrift/gRPC

PROFESSIONAL EXPERIENCE

Meta

United Kingdom

Production Engineer, Network AI PE

September 2025 – Present

- **Drain Monitoring & Observability:** Built network data collections to identify the drain state of devices across a fleet of 100,000+ switches, validating BGP, switch agent, and other config application mediums across different device and switch roles. Developed large-scale data processing and auditing pipelines; designed alert infrastructure that scales to millions of network interfaces across the fleet.
- **NVLINK Bug Discovery & Automated Repair:** Rootcaused a configuration misapplication affecting >25% of the NVLINK switch fleet (~40,000 devices) using AI-assisted log parsing. Devised an automated mechanism to detect impacted switches and apply targeted repair fleet-wide.
- **CEL Compiler CLI Tool (Go):** Extended a Go CLI tool that parsed CEL (Common Expression Language) configuration files and compiled corresponding device domain filters. Enabled extensive dev-time testing across all teams consuming the config, eliminating the need for fleet-wide canary deployments to validate changes.

- **Production Service Migration:** Migrated APIs for a network drain service handling >1 million daily requests; implemented A/B testing to guarantee zero user traffic impact during transition.
- **Oncall – Network AI PE:** Joined team oncall within 1 month of onboarding; monitored a 100,000+ device fleet and responded to live incidents. Developed rootcause analysis skills distinguishing noise from symptoms; deepened knowledge of OS internals (kernel and system service behaviour) and network protocols (BGP, DHCP, ZTP).

Zilliqa

London, United Kingdom

Intern, Software Engineering

July – September 2023

- Modified the Ethereum block explorer Otterscan to work with a Zilliqa node using React and TypeScript, contributing to Zilliqa's official block explorer development. This tool is set to be Zilliqa's official block explorer.
- Worked with Zilliqa core team on Ethereum Virtual Machine APIs written in C++ to enable cross-chain interoperability.
- Developed a DevOps tool for node building using Rust and worked with Google Cloud VMs and AWS.

PROJECTS

Pronto – Federated Kubernetes Scheduler

Go, Kubernetes Scheduling Framework, gRPC, Docker, FPCA github.com/LucaChot/pronto-framework

- Designed and implemented a custom Kubernetes scheduler plugin using the official Scheduling Framework API, replacing the default scoring and filtering logic with a signal-based placement algorithm.
- Applied Functional PCA (FPCA) across a fleet of worker nodes to distil real-time CPU and memory telemetry into a scalar load signal, enabling the scheduler to route pods to nodes with genuine available capacity rather than relying on declared resource requests.
- Built a distributed aggregation layer using gRPC streaming, where per-node FPCA estimates are continuously merged into a shared global embedding — a form of federated learning applied to infrastructure load prediction.
- Evaluated against the default kube-scheduler across CPU-intensive and ML inference workloads, demonstrating improved node utilisation balance and reduced job completion times under batch workloads.

HOBBIES AND INTERESTS

**Halliwick Penguin SC Swimming Pool
Volunteer Trainer**

London, United Kingdom

September 2017 - July 2018

- Assisted individuals with disabilities in learning to swim, ensuring safety and engagement in the sessions.

Other: Bilingual (English / Italian), travel (driving license holder), hiking, yoga, working out, football, skiing/snowboarding.